



Introduction to Information and Communication Technology (ICT)

GRADE 10 · MODULE 1.1

This module introduces the fundamental concepts of ICT – data, information, systems, and the role of ICT in today's world.

1

Define Data & Information

2

Understand Systems

3

Identify System Components

4

Explain ICT Importance

What is Data?

"The numbers, words, images and symbols which do not bear a meaning, when standing alone are called data."

Raw Facts

Data has no meaning by itself — it needs processing to become useful.

Examples

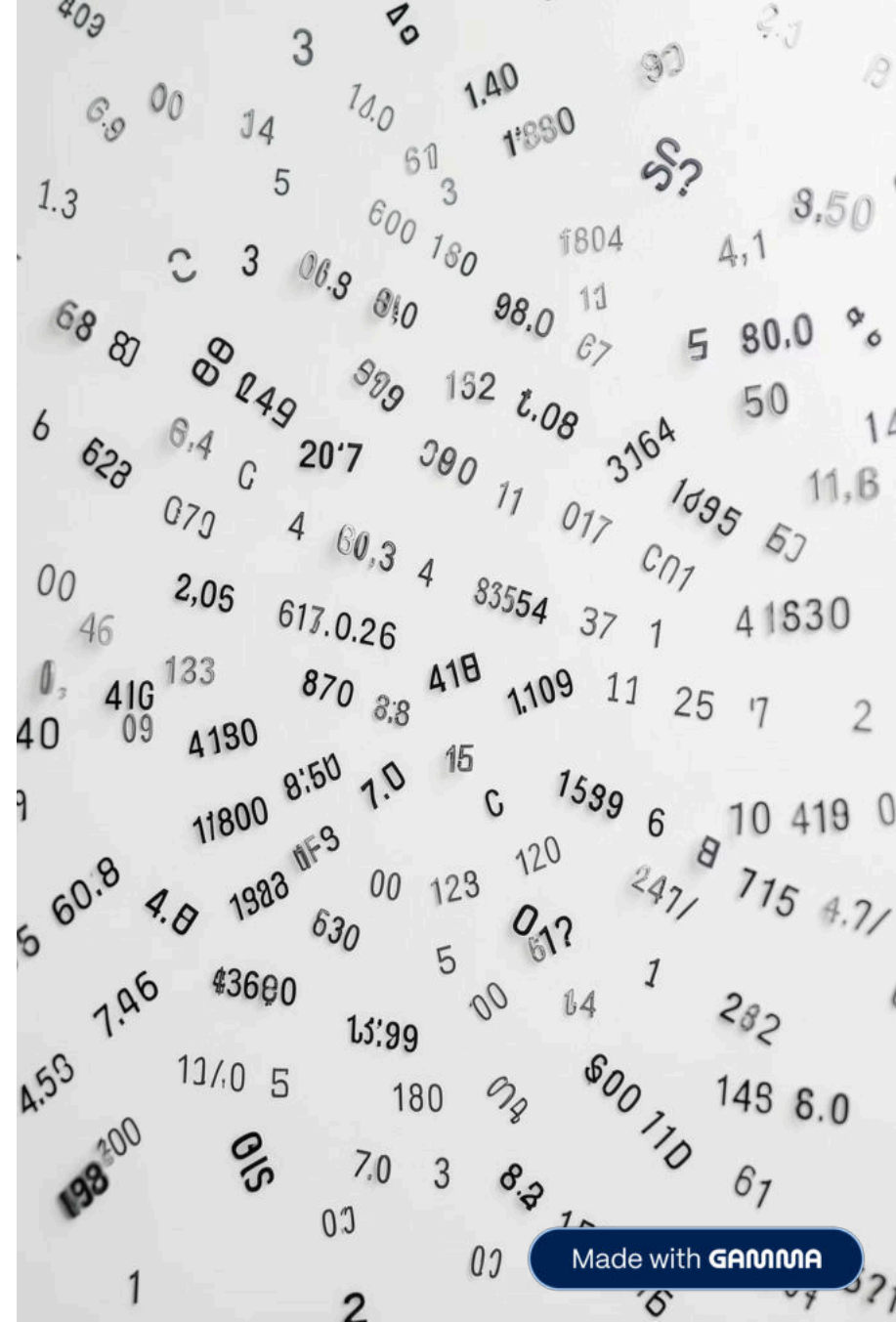
Numbers: **78, 90, 79, 67, 76, 98**

Names: **Ravi, Saman, Rizwan**

Types of Data

Numbers, text, images, and symbols — all are forms of raw data.

Activity: Provide 3 more examples of data from daily life (e.g., temperature readings, barcode numbers, student ID numbers).



What is Information?

"Meaningful information can be arrived at by arranging and processing data, which can be used to make decisions."

Data (Raw)

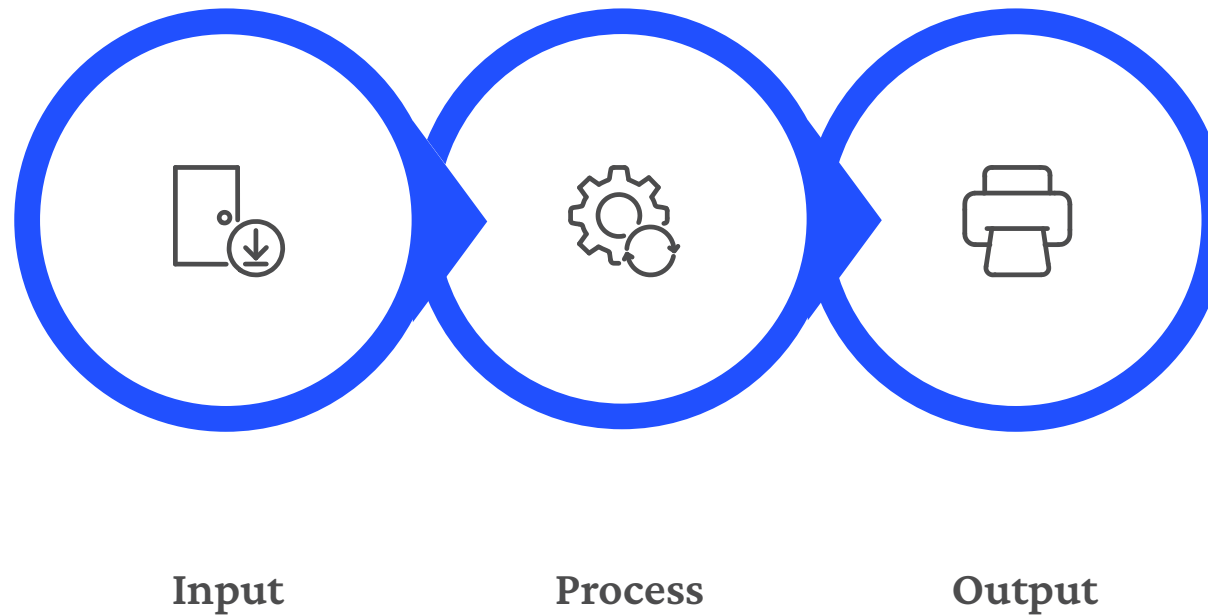
- 78, 90, 79, 67, 76, 98
- Individual marks only
- No meaning alone

Information (Processed)

- Ravi scored 468 total, ranked 2nd
- Student performance summary
- Helps teacher make decisions

Information is **processed, organized, and structured data** that is useful for decision-making. When totals are calculated, averages found, and ranks assigned — data becomes information.

Input – Process – Output Concept



Every computer system follows this fundamental flow: data enters as **input**, is **processed** through operations or calculations, and produces **output** as meaningful information.

ATM Transaction Example

Card insertion (**Input**) → PIN verification (**Process**) → Cash withdrawal (**Output**)

Student Marks Example

Raw marks entered (**Input**) → Totals & averages calculated (**Process**) → Results table produced (**Output**)

What is a System?

"A system is a combination of components that work together to fulfil a task."



Input

Data entered into the system



Processing

Operations performed on data



Output

Information produced



Storage

Keeping data/information for future use

A computer is called a "**System**" because it processes data we input according to commands and provides required information. Real-world examples include **ATM machines**, **fingerprint attendance systems**, and **QR code readers**.



Characteristics of Quality Information

Not all information is suitable for making decisions. Information obtained should be of **good quality**.

Characteristic	Explanation	Example
Relevancy	Only include necessary information	Not all Grade 1 records needed – only highest qualification
Completeness	Information must be thorough	Small sample for Per Capita Income would be incomplete
Accuracy	Information must be correct	Wrong patient health info could be harmful
Timeliness	Information must be up-to-date	Today's weather report may not suit tomorrow's decisions
Cost Effectiveness	Cost of collecting info should not exceed benefits	Spending more to collect info than profits gained is a loss

 **Activity:** Write one additional real-life example for each characteristic of quality information.

ICT Definition and Scope

"Technology used in various ways to convert data into information and then to exchange them is called Information and Communication Technology (ICT)."

Information Technology (IT)

Processing data to produce meaningful information using computers and software systems.

Communication Technology

Exchanging information between people and systems using networks, internet, and digital devices.

Humans have processed and exchanged information since ancient times. Today, technology has made this process faster and easier than ever before. ICT combines both components — **data processing** and **information exchange** — into one powerful field.

Brief History of Computing

Computing has evolved over **5,000+ years**, from simple counting tools to complex electronic machines. **Charles Babbage** is known as the **Father of Computing**.

Year	Invention	Inventor
~5000 years ago	Abacus	Unknown
1642	Adding Machine	Blaise Pascal
1674	Improved Calculator (× & ÷)	Gottfried Wilhelm Von Leibnitz
~1800s	Mechanical Loom (Punch Cards)	Joseph Jacquard
1800s	Analytical Engine	Charles Babbage
1800s	First Programmer	Ada Augusta Lovelace
1944	MARK 1 (Automatic Sequence Control Calculator)	Howard Aiken

Computer Generations Overview

All early computers were mechanical. From the time computers became automated, they were categorized into **generations** based on their hardware technology.

1st Generation (1940–1956)

Hardware: Vacuum Tubes, Punch Cards

Software: Machine Language, Assembly Language

Characteristics: High heat, slow, large, expensive, non-portable, high electricity consumption

Key Computers: ENIAC, EDVAC, EDSAC, UNIVAC, IBM 701

2nd Generation (1956–1963)

Hardware: Transistors, Tape, Floppy Disks

Software: High-level Programming Languages, Assembly Language

Characteristics: Smaller, less heat, lower power, faster, still expensive

Key Computers: Honeywell 400, IBM 7030, CDC 1604, UNIVAC LARC

Module 1.1 Summary

1 Data → Process → Information → Decisions

2 Computer is a system: Input – Process – Output – Store

3 Quality information must be: Relevant, Complete, Accurate, Timely, Cost-effective

4 ICT = Information Technology + Communication Technology

5 Computing evolution spans 5,000+ years

Review Questions:

1. What is the difference between data and information?
2. Name the components of a system
3. List 3 characteristics of quality information
4. Who is called the Father of Computing?
5. Define ICT in your own words

 **Group Activity:** Find information on computer history and make a report as a group activity.